Project: Worked examples
Title: Psychrometrics
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1. Scope and method

Calculates the state of moist air from dry-bulb and one other property at a stated pressure or altitude. It then applies one air-handling process with its heat and moisture balance: sensible heating or cooling, dehumidification, humidification or mixing.

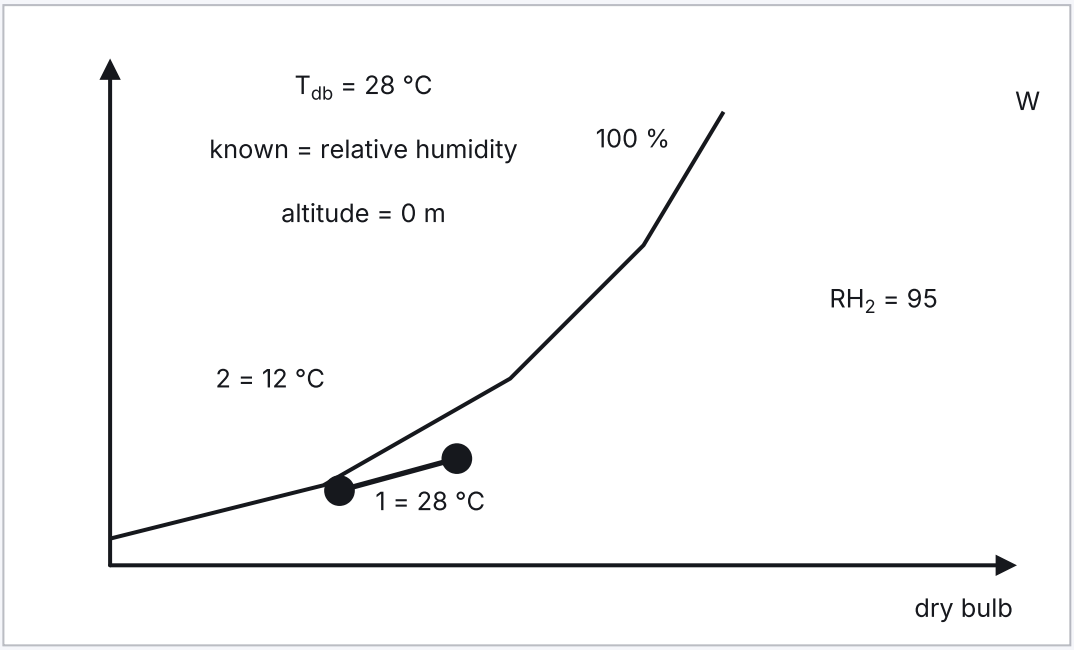


Figure 1 — Cooling with dehumidification: the state cools to T_2 at RH_2, losing moisture. Not to scale; the labels are the inputs as entered.

WHAT IT CALCULATES	RELATION
Saturation and humidity	$p_{ws}(T)$ by Hyland–Wexler over water at or above 0 °C and over ice below; $W = 0.621945\ p_w/(p - p_w)$; $\varphi = p_w/p_{ws}$
Enthalpy and volume per kilogram of dry air	$h = 1.006\ T + W(2501 + 1.86\ T)$ kJ/kg; $v = 0.287042\ (T + 273.15)(1 + 1.607858\ W)/p$
Dew point and wet-bulb temperatures, by bisection	$p_{ws}(T_{dp}) = p_w$; the wet-bulb from the adiabatic-saturation relation
Sensible heating or cooling at constant moisture content	$\Phi = \dot{m}\ (h_2 - h_1)$
Cooling with dehumidification, less the heat leaving in the condensate	$Q_{total} = \dot{m}(h_1 - h_2) - \dot{m}_w\ h_w$, with $\dot{m}_w = \dot{m}(W_1 - W_2)$
Sensible and latent split of the coil load, along $T_1 \rightarrow T_2$ at W_1 then $W_1 \rightarrow W_2$ at T_2	$Q_{sensible} = \dot{m}(h_1 - h(T_2, W_1))$; $Q_{latent} = \dot{m}(h(T_2, W_1) - h_2)$; $SHR = Q_{sensible}/(Q_{sensible} + Q_{latent})$; with RH_2 blank and T_2 above the entering dew point the coil is dry, $W_2 = W_1$
Steam and adiabatic humidification	$h_2 = h_1 + (W_2 - W_1)\ h_{steam}$ for steam; $h_2 = h_1$ for adiabatic (spray) humidification

WHAT IT CALCULATES	RELATION
Mixing two airstreams, weighted by dry-air mass flow	$W_{mix} = \Sigma \dot{m}\ W/\Sigma \dot{m}$; $h_{mix} = \Sigma \dot{m}\ h/\Sigma \dot{m}$

Basis: Psychrometrics (ideal-gas relations, processes)[1](Ch. 1); Saturation pressure of water and ice[2]; Properties of humid air (principles)[3](§1).

2. Inputs

SYMBOL	DESCRIPTION	VALUE	UNIT	SOURCE	VALID RANGE
known	Second known property	relative humidity		assumption: RH given.	any
process	Process	cooling with dehumidification		assumption: Chilled-water coil.	any
altitude	Altitude (used when no pressure is entered)	0	m	assumption: Sea-level site.	-500 ... 5000 m
T_{db}	Dry-bulb temperature	28	°C	spec: CIBSE Guide A summer design condition for the site, assumed	-40 ... 60 °C
RH	Relative humidity	60		spec: Design condition	5 ... 100
m_{da}	Dry-air mass flow	2.4	kg/s	spec: AHU schedule AHU-01: 2.0 m³/s ≈ 2.4 kg/s dry air	0 ... 1000 kg/s
h_{steam}	Steam enthalpy (steam humidification)	2676		assumption: Steam humidification only; this cooling coil does not use it, so the template default is kept.	2500 ... 3000
T_2	Leaving dry-bulb (sensible, cooling)	12	°C	spec: AHU schedule AHU-01 off-coil 12 °C	-40 ... 80 °C
RH_2	Leaving relative humidity (cooling, evaporative)	95		assumption: Off-coil 95 % RH (typical contact factor).	5 ... 100

3. Assumptions

A1. Moist air is an ideal mixture of ideal gases. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A2. The enhancement factor is taken as 1. (ASHRAE Fundamentals Ch. 1 ideal-gas relations.)
A3. Condensate leaves at the coil leaving dry-bulb. (Process conventions.)
A4. Steam adds the enthalpy h_{steam} at negligible mass of dry air. (Process conventions.)
A5. Adiabatic humidification follows a line of constant enthalpy, which is close to constant wet-bulb. (Evaporative process idealisation.)
A6. known (Second known property) is taken as relative humidity. (RH given.)
A7. process (Process) is taken as cooling with dehumidification. (Chilled-water coil.)
A8. altitude (Altitude (used when no pressure is entered)) is taken as 0 m. (Sea-level site.)

A9. h_{steam} (Steam enthalpy (steam humidification)) is taken as 2676. (Steam humidification only; this cooling coil does not use it, so the template default is kept.)
A10. RH_2 (Leaving relative humidity (cooling, evaporative)) is taken as 95. (Off-coil 95 % RH (typical contact factor).)

4. Working

[2] p_{atm} = 101.325 (1 - 2.25577e-5 z)^5.2559 [1](Ch. 1 eq. 3)
= z = 0 m
= 101.32 kPa
Barometric pressure (standard atmosphere)
[6] $p_{ws,1}$ = p_{ws}(t) Hyland-Wexler (entering)
= t = 28 °C
= 3.7822 kPa
Saturation pressure, state 1
[7] $p_{w,1}$ = p W/(0.621945 + W)
= W = 0.014249
= 2.2693 kPa
Vapour pressure, state 1
[8] $T_{1,out}$ = dry-bulb (entering)
= t = 28 °C
= 28 degC
Dry-bulb, state 1
[9] W_1 = 0.621945 p_w/(p - p_w)
= 2.2693 / (101.32 - 2.2693)
= 14.249
Humidity ratio, state 1 (g/kg dry air)
[10] RH_1 = p_w / p_{ws}
= 2.2693 / 3.7822
= 60
Relative humidity, state 1 (%)
[11] φ_1 = p_w / p_{ws}
= 2.2693 / 3.7822
= 60 %
Relative humidity, state 1, as a percentage quantity (the same value as RH_1; a plain-number output cannot be linked to a percentage input)
[12] $T_{dp,1}$ = p_{ws}(t_{dp}) = p_w, bisection
= p_w = 2.2693 kPa
= 19.514 degC
Dew point, state 1
[13] $T_{wb,1}$ = adiabatic-saturation relation, bisection
= t = 28, W = 0.014249
= 22.074 degC
Wet-bulb, state 1
[14] h_1 = 1.006 t + W (2501 + 1.86 t)
= 1.006 × 28 + 0.014249 × (2501 + 1.86 × 28)
= 64.546
Enthalpy, state 1 (kJ/kg dry air)
[15] v_1 = 0.287042 (t + 273.15)(1 + 1.607858 W)/p
= p = 101.32 kPa
= 0.87267 m³/kg
Specific volume, state 1 (per kg dry air)
[16] ρ_1 = (1 + W)/v
= (1 + 0.014249) / 0.87267
= 1.1622 kg/m³
Moist-air density, state 1

[21] $p_{ws,2} = p_{ws}(t)$ Hyland-Wexler (leaving)
= t = 12 °C
= 1.4026 kPa
Saturation pressure, state 2
[22] $p_{w,2} = p\ W/(0.621945 + W)$
= W = 0.0082878
= 1.3325 kPa
Vapour pressure, state 2
[23] $T_{2,out}$ = dry-bulb (leaving)
= t = 12 °C
= 12 degC
Dry-bulb, state 2
[24] $W_2 = 0.621945\ p_w/(p - p_w)$
= 1.3325 / (101.32 − 1.3325)
= 8.2878
Humidity ratio, state 2 (g/kg dry air)
[25] $RH_2 = p_w / p_{ws}$
= 1.3325 / 1.4026
= 95
Relative humidity, state 2 (%)
[26] $\varphi_2 = p_w / p_{ws}$
= 1.3325 / 1.4026
= 95 %
Relative humidity, state 2, as a percentage quantity (the same value as RH_2; a plain-number output cannot be linked to a percentage input)
[27] $T_{dp,2} = p_{ws}(t_{dp}) = p_w$, bisection
= p_w = 1.3325 kPa
= 11.225 degC
Dew point, state 2
[28] $T_{wb,2}$ = adiabatic-saturation relation, bisection
= t = 12, W = 0.0082878
= 11.553 degC
Wet-bulb, state 2
[29] $h_2 = 1.006\ t + W\ (2501 + 1.86\ t)$
= 1.006 × 12 + 0.0082878 × (2501 + 1.86 × 12)
= 32.985
Enthalpy, state 2 (kJ/kg dry air)
[30] $v_2 = 0.287042\ (t + 273.15)(1 + 1.607858\ W)/p$
= p = 101.32 kPa
= 0.81856 m³/kg
Specific volume, state 2 (per kg dry air)
[31] $\rho_2 = (1 + W)/v$
= (1 + 0.0082878) / 0.81856
= 1.2318 kg/m³
Moist-air density, state 2
[32] $m_w = \dot{m}\ (W_1 - W_2)$
= 2.4 × (14.249 − 8.2878) g/kg
= 0.014306 kg/s
Condensate
[33] $h_w = 4.186\ t_2$ (condensate enthalpy)
= 4.186 × 12
= 50.232
Condensate enthalpy, kJ/kg

[34] $Q_{total} = \dot{m} (h_1 - h_2) - \dot{m}_w\, h_w$
$= 2.4 \times (64.546 - 32.985) - 0.014306 \times 50.232$
$= 75.027$ kW
Total cooling load
[35] $Q_{sensible} = \dot{m} (h_1 - h(t_2, W_1))$
$= 2.4 \times (64.546 - 48.026)$
$= 39.648$ kW
Sensible part: the air cooled from t_1 to t_2 at the entering humidity ratio W_1
[36] $Q_{latent} = \dot{m} (h(t_2, W_1) - h_2)$
$= 2.4 \times (48.026 - 32.985)$
$= 36.098$ kW
Latent part: the moisture removed from W_1 to W_2 at t_2
[37] $SHR = Q_{sensible} / (Q_{sensible} + Q_{latent})$
$= 39.648 / (39.648 + 36.098)$
$= 0.52344$
Sensible heat ratio

5. Results

SYMBOL	VALUE	DESCRIPTION
$T_{1,out}$	28 °C	Dry-bulb temperature, state 1
φ_1	60 %	Relative humidity, state 1, as a quantity in % (for linking)
$T_{dp,1}$	19.51 °C	Dew point, state 1
$T_{wb,1}$	22.07 °C	Wet-bulb temperature, state 1
W_1	14.249	Humidity ratio, state 1, in grams per kilogram of dry air
h_1	64.546	Enthalpy, state 1, in kilojoules per kilogram of dry air
$T_{2,out}$	12 °C	Dry-bulb temperature, state 2
W_2	8.2878	Humidity ratio, state 2, in grams per kilogram of dry air
h_2	32.985	Enthalpy, state 2, in kilojoules per kilogram of dry air
$p_{ws,1}$	3.782 kPa	Saturation pressure, state 1
$p_{w,1}$	2.269 kPa	Vapour pressure, state 1
RH_1	60	Relative humidity, state 1, as a number from 0 to 100
v_1	0.8727 m³/kg	Specific volume, state 1
ρ_1	1.162 kg/m³	Moist-air density, state 1
$p_{ws,2}$	1.403 kPa	Saturation pressure, state 2
$p_{w,2}$	1.332 kPa	Vapour pressure, state 2
RH_2	95	Relative humidity, state 2, as a number from 0 to 100
φ_2	95 %	Relative humidity, state 2, as a quantity in % (for linking)
$T_{dp,2}$	11.22 °C	Dew point, state 2
$T_{wb,2}$	11.55 °C	Wet-bulb temperature, state 2
v_2	0.8186 m³/kg	Specific volume, state 2
ρ_2	1.232 kg/m³	Moist-air density, state 2
m_w	0.01431 kg/s	Condensate

SYMBOL	VALUE	DESCRIPTION
Q_{total}	75.03 kW	Total cooling
$Q_{sensible}$	39.65 kW	Sensible cooling
Q_{latent}	36.10 kW	Latent cooling
SHR	0.52344	Sensible heat ratio

6. Checks (governing first)

None.

7. Applicability

None.

8. Verification

`Template bs.psychrometrics@1.4.0 Content hash 9ffd235faa1c3dae Engine calc-engine/1.0.0`
`Known-answer cases passed 4/4 Test run 2026-10-23 12:00 UTC TempestOS 0.26.0`

9. References

1. American Society of Heating, Refrigerating and Air-Conditioning Engineers. ASHRAE Handbook—Fundamentals. ASHRAE.
2. Hyland RW, Wexler A. Formulations for the thermodynamic properties of the saturated phases of H2O from 173.15 K to 473.15 K. ASHRAE Trans. 1983;89(2A):500-519.
3. Chartered Institution of Building Services Engineers. CIBSE Guide C: Reference Data. CIBSE; 2007.